

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

LABORATORY INVESTIGATION & FORENSIC EVIDENCE REPORT

“Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training”

Lab Title: SBT-DF203 Lab 4: SMTP Email Traffic Forensics

Course Name: Basic Networking Skills for Digital Forensics

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Operating Platform: Kali Linux 2026 (Rolling Release)

Date of Submission: 14/09/2026

Investigation Target: Historical SMTP Packet Stream (smtp.pcap)

Report Version: Version 1.0 (Final Release)

Executive Summary

This report documents the forensic analysis of a historical SMTP packet capture (smtp.pcap). The investigation identified a complete email exchange between a client (10.10.1.4:1470) and a server (74.53.140.153:25, Exim 4.69). The SMTP conversation was reconstructed, including the authentication exchange (AUTH LOGIN), envelope fields (MAIL FROM, RCPT TO), and message content. Base64-encoded credentials were decoded offline with the password masked per policy. The email contained a multipart message with plain text, HTML, and a NEWS.txt attachment. Critically, although the server advertised STARTTLS, the client did not use it, resulting in the entire session being transmitted in plaintext, exposing credentials and message content.

1. Evidence Integrity and Scope

- Original Capture Evidence:** evidence/smtp.pcap (Preserved read-only bitstream capture file).
- Working Forensic Duplicate:** working/smtp_working.pcap (Bit-identical working copy used for analysis).
- Cryptographic Hash (SHA-256):** 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab (Both files match perfectly).
- Capture Metadata:** 60 packets, 9.198384 seconds duration, 27 kB file size, Ethernet encapsulation, microsecond precision.

2. SMTP Timeline

The following chronological timeline traces the complete application-layer transaction between client and server:

Frame	Time	Dir	Command / Code	Details
6	02:06:08.219	S → C	220	xc90.websitewelcome.com ESMTP Exim 4.69
7	02:06:08.224	C → S	EHLO	EHLO GP
9	02:06:08.566	S → C	250	SIZE 52428800, PIPELINING, AUTH PLAIN LOGIN, STARTTLS
10	02:06:08.568	C → S	AUTH LOGIN	Client initiates interactive authentication

11	02:06:08.911	S → C	334	VXNlcm5hbWU6 (Username challenge)
12	02:06:08.911	C → S	(Base64)	Z3VycGFydGFwQHBhdHJpb3RzLmlu
13	02:06:09.253	S → C	334	UGFzc3dvcmQ6 (Password challenge)
14	02:06:09.254	C → S	(Base64)	cHVuamFiQDEyMw==
15	02:06:09.613	S → C	235	Authentication succeeded
16	02:06:09.614	C → S	MAIL FROM	<gurpartap@patriots.in>
18	02:06:09.957	C → S	RCPT TO	<raj_deol2002in@yahoo.co.in>
20	02:06:10.319	C → S	DATA	Client requests to begin message transmission
21	02:06:10.661	S → C	354	Enter message, ending with "." on a line by itself
52	02:06:12.248	S → C	250	OK id=1Mugho-0003Dg-Un (Message accepted)
54	02:06:14.763	C → S	QUIT	Client terminates transaction
56	02:06:15.105	S → C	221	xc90.websitewelcome.com closing connection

3. Authentication Evidence and Redaction

- **Authentication Mechanism:** AUTH LOGIN (Challenge-Response SASL mechanism over plaintext TCP).
- **Username Challenge & Response:** Server Challenge: 334 VXNlcm5hbWU6 ('Username:') | Client Response: Z3VycGFydGFwQHBhdHJpb3RzLmlu
- **Password Challenge & Response:** Server Challenge: 334 UGFzc3dvcmQ6 ('Password:') | Client Response: cHVuamFiQDEyMw==
- **Decoded Username:** gurpartap@patriots.in
- **Decoded Password (Redacted):** Masked as p*****3 per ICDFA forensic reporting policy.
- **Redaction Methodology:** Processed offline via local Python script (scripts/decode_base64.py). No external or cloud decoders utilized.

```
import base64
u_b64 = 'Z3VycGFydGFwQHBhdHJpb3RzLmlu'
p_b64 = 'cHVuamFiQDEyMw=='
username = base64.b64decode(u_b64).decode('utf-8')
password = base64.b64decode(p_b64).decode('utf-8')
masked_pw = password[0] + '*' * (len(password) - 2) + password[-1]
print(f'Username: {username}')
print(f'Password: {masked_pw} (masked for reporting)')
```

4. Email Reconstruction

Reconstruction of TCP stream 0 reveals the complete MIME multipart hierarchy, message headers, body parts, and raw attachment:

- **From:** "Gurpartap Singh" <gurpartap@patriots.in>

- **To:** <raj_deol2002in@yahoo.co.in>
- **Subject:** SMTP
- **Date:** Mon, 5 Oct 2009 11:36:07 +0530
- **Message-ID:** <000301ca4581\$ef9e57f0\$cedb07d0\$@in>
- **MIME-Version:** 1.0
- **Content-Type:** multipart/mixed; boundary="-----_NextPart_000_0004_01CA45B0.095693F0"
- **X-Mailer:** Microsoft Office Outlook 12.0

Payload Breakdown: Multipart Body Structure:

- **Part 1 (text/plain):** Content: "Hello / I send u smtp pcap file / Find the attachment / GPS"
- **Part 2 (text/html):** HTML formatted rendering generated by Microsoft Word 12 (Font definitions: Cambria Math, Calibri).
- **Part 3 (Attachment):** Filename: NEWS.txt (MIME Content-Type: text/plain, containing comprehensive Dev-C++ software release history notes from v4.9.4.1 up to v4.9.9.1).

5. Network Metadata

Layer 2 and Layer 3 addressing captured during the SMTP session:

Field	Client Node	Server Node
IP Address (Layer 3)	10.10.1.4	74.53.140.153
TCP Port (Layer 4)	1470 (Ephemeral)	25 (Standard SMTP)
MAC Address (Layer 2)	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2

6. Encryption Assessment and Limitations

- **STARTTLS Advertisement:** Server response 250 in Frame 9 advertised STARTTLS capability.
- **Client Opportunistic Encryption Failure:** The client completely ignored the STARTTLS advertisement and immediately issued AUTH LOGIN in Frame 10.
- **TLS Traffic Volume:** 0 TLS / SSL encrypted packets detected throughout the entire capture.
- **Forensic Vulnerability:** Because opportunistic TLS upgrade was not initiated, the entire session (including user authentication credentials, envelope sender/recipient metadata, message headers, email body, and attachment data) was exposed in unencrypted plaintext.
- **Forensic Limitations:** Base64 encoding is an ASCII serialization scheme, NOT encryption; it provides zero confidentiality. Capture also contains extraneous ICMP fragmentation-needed packets (Frames 26–30) originating from 192.168.1.1, unrelated to the SMTP stream.

7. Findings Worksheet

Consolidated forensic findings answering core investigative questions:

Investigative Question	Forensic Finding
Session start/end timestamps	2009-10-05 02:06:07.492060 → 02:06:16.690444 UTC
Client IP / MAC / Port	10.10.1.4 / 00:1f:33:d9:81:60 / TCP 1470
Server IP / MAC / Port	74.53.140.153 / 00:e0:1c:3c:17:c2 / TCP 25
Server banner identification	xc90.websitewelcome.com ESMTP Exim 4.69 #1
Client software user-agent	Microsoft Office Outlook 12.0
Authentication mechanism	AUTH LOGIN (Base64 encoded, unencrypted)
Envelope sender (MAIL FROM)	gurpartap@patriots.in
Envelope recipient (RCPT TO)	raj_deol2002in@yahoo.co.in
Subject line	SMTP
MIME structure & Body types	multipart/mixed (contains text/plain and text/html)
Attachment filename	NEWS.txt (Dev-C++ version history)
STARTTLS observed in transit	No (Advertised by server, but never invoked by client)

8. Screenshots and Evidence

The following 24 figures depict the end-to-end evidence processing, packet inventory, Base64 credential decoding, full MIME reconstruction, attachment carving, and TLS assessment:

```
kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
(kali@kali)~$ cd ~
$ cd ~
mkdir -p ~/SBT-DF203-Lab4/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab4

(kali@kali)~/SBT-DF203-Lab4$ pwd
/home/kali/SBT-DF203-Lab4
(kali@kali)~/SBT-DF203-Lab4$ find . -maxdepth 1 -type d -print
./
./exported
./evidence
./working
./screenshots
./scripts
./reports

(kali@kali)~/SBT-DF203-Lab4$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.7 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [113 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [261 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 76.7 MB in 38s (2,030 kB/s)
2205 packages can be upgraded. Run 'apt list --upgradable' to see them.
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
  gir1.2-girepository-2.0  libcrypt-dev  libkmlbase1t64  libmplex2-2.1-0t64  libqt6test6  python3-fs  rpcsvc-proto
  libann0  libgav1-1  libkmlcore1t64  libpoppler147  libqt6xml6  python3-gpg  samba-ad-dc
  libarmadillo14  libgdal37  libkmlengine1t64  libportmidi0  libraw23t64  python3-pyqt6.sip  samba-ad-provision
  libavc1394-0  libgirepository-1.0-1  liblab-gamut1  libpython3.13-dev  libwireshark18  python3.13  samba-dsdb-modules
  libblosc2-4  libpgmep6t64  libmbedcrypto16  libqt6openglwidgets6  libwiretap15  python3.13-dev  python3.13-minimal
  libcdt5  libjs-jquery-ui  libmjpegutils-2.1-0t64  libqt6sql6  libwsutil16  python3.13-venv
  libcgraph6  libjs-underscore  libmpeg2encpp-2.1-0t64  libqt6sql6-sqlite  libzxing3
Use 'sudo apt autoremove' to remove them.

Upgrading:
```

Figure 1: Workspace Directory Tree & Tool Verification — Initializing /SBT-DF203-Lab4 hierarchy and verifying Wireshark, TShark, Python3, and Wget.

```
kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

(kali@kali)~[~/SBT-DF203-Lab4]
$ cd ~/SBT-DF203-Lab4
$ wget -O evidence/sntp.pcap 'https://wiki.wireshark.org/uploads/_mon_import/_attachments/SampleCaptures/sntp.pcap'
--2026-09-13 17:39:35-- https://wiki.wireshark.org/uploads/_mon_import/_attachments/SampleCaptures/sntp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org) ... 104.26.10.240, 104.26.11.240, 172.67.75.39, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|104.26.10.240|:443 ... connected.
HTTP request sent, awaiting response ... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/_mon_import/_attachments/SampleCaptures/sntp.pcap [following]
--2026-09-13 17:39:38-- https://gitlab.com/wireshark/wireshark/-/wikis/uploads/_mon_import/_attachments/SampleCaptures/sntp.pcap
Resolving gitlab.com (gitlab.com) ... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443 ... connected.
HTTP request sent, awaiting response ... 404 Not Found
2026-09-13 17:39:39 ERROR 404: Not Found.

(kali@kali)~[~/SBT-DF203-Lab4]
$ ls -lh evidence/sntp.pcap
file evidence/sntp.pcap
-rw----- 1 kali kali 28K Sep 13 17:45 evidence/sntp.pcap
evidence/sntp.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 65535)

(kali@kali)~[~/SBT-DF203-Lab4]
$ cd ~/SBT-DF203-Lab4
$ cp --preserve=timestamps evidence/sntp.pcap working/sntp_working.pcap
$ sha256sum evidence/sntp.pcap working/sntp_working.pcap | tee reports/sntp_capture_hashes.txt
cat reports/sntp_capture_hashes.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/sntp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/sntp_working.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/sntp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/sntp_working.pcap

(kali@kali)~[~/SBT-DF203-Lab4]
$ capinfos evidence/sntp.pcap | tee reports/sntp_capinfos.txt
cat reports/sntp_capinfos.txt
File name: evidence/sntp.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 60
File size: 27 kB
Data size: 26 kB
```

Figure 2: Evidence Ingestion, File Verification & SHA-256 Hashing — Ingesting sntp.pcap, verifying libpcap format (28K), generating working copy, and computing matching SHA-256 digests.

```
kali@kali: ~/SBT-DF203-Lab4

Session Actions Edit View Help

Earliest packet time: 2009-10-05 02:06:07.492060
Latest packet time: 2009-10-05 02:06:16.690444
Data byte rate: 2,920 bytes/s
Data bit rate: 23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256: 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1: 3def1a1fed849e7f23b66e6925ddff05845a400b
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 60
File name: evidence/smtp.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 60
File size: 27 kB
Data size: 26 kB
Capture duration: 9.198384 seconds
Earliest packet time: 2009-10-05 02:06:07.492060
Latest packet time: 2009-10-05 02:06:16.690444
Data byte rate: 2,920 bytes/s
Data bit rate: 23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256: 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1: 3def1a1fed849e7f23b66e6925ddff05845a400b
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 60
```

Figure 3: *Capinfos Capture File Metadata Dissection* — Detailed capinfos output: 60 packets, 9.198s duration, 23 kbps data rate, and Ethernet encapsulation.

```
kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

(kali@kali) [~/SBT-DF203-Lab4]
$ cd ~/SBT-DF203-Lab4
tshark -r working/sntp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt
cat reports/tcp_conversations.txt
Warning: program compiled against libxml 215 using older 214

TCP Conversations
Filter:<No Filter>

| Frames | Bytes | | Frames | Bytes | | Total | Relative | Duration |
|-----|-----| |-----|-----| |-----|-----|-----|
10.10.1.4:1470 <--> 74.53.140.153:25 | 25 | 1,980 bytes | 28 | 22 kB | 53 | 24 kB | 0.036986000 | 7.5777

TCP Conversations
Filter:<No Filter>

| Frames | Bytes | | Frames | Bytes | | Total | Relative | Duration |
|-----|-----| |-----|-----| |-----|-----|-----|
10.10.1.4:1470 <--> 74.53.140.153:25 | 25 | 1,980 bytes | 28 | 22 kB | 53 | 24 kB | 0.036986000 | 7.5777

(kali@kali) [~/SBT-DF203-Lab4]
$ tshark -r working/sntp_working.pcap -Y 'smtp' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info | tee reports/sntp_packet_inventory.tsv
cat reports/sntp_packet_inventory.tsv
Warning: program compiled against libxml 215 using older 214
6 2009-10-05T02:06:08.219663000-0400 74.53.140.153 25 10.10.1.4 1470 S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 | We do not authorize the use of this system to transport unsolicited, and/or bulk e-mail.
7 2009-10-05T02:06:08.224809000-0400 10.10.1.4 1470 74.53.140.153 25 C: EHLO GP
9 2009-10-05T02:06:08.566183000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] |
SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP
10 2009-10-05T02:06:08.568729000-0400 10.10.1.4 1470 74.53.140.153 25 C: AUTH LOGIN
11 2009-10-05T02:06:08.911081000-0400 74.53.140.153 25 10.10.1.4 1470 S: 334 VXNlcm5hbWU6
12 2009-10-05T02:06:08.911655000-0400 10.10.1.4 1470 74.53.140.153 25 C: User: Z3VycGFydGFwQHhhdHJpb3RzLmlu
13 2009-10-05T02:06:09.253544000-0400 74.53.140.153 25 10.10.1.4 1470 S: 334 UGFzc3dvcmQ6
14 2009-10-05T02:06:09.254118000-0400 10.10.1.4 1470 74.53.140.153 25 C: Pass: cHVuamF1QDEyMw==
15 2009-10-05T02:06:09.613798000-0400 74.53.140.153 25 10.10.1.4 1470 S: 235 Authentication succeeded
16 2009-10-05T02:06:09.614414000-0400 10.10.1.4 1470 74.53.140.153 25 C: MAIL FROM: <gurpartap@patriots.in>
17 2009-10-05T02:06:09.956765000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 OK
18 2009-10-05T02:06:09.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19 2009-10-05T02:06:10.319708000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 Accepted
20 2009-10-05T02:06:10.320203000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA
21 2009-10-05T02:06:10.661679000-0400 74.53.140.153 25 10.10.1.4 1470 S: 354 Enter message, ending with "." on a line by itself
```

Figure 4: TCP Conversation Summary & Initial SMTP Packet Inventory — Mapping TCP session 10.10.1.4:1470 <-> 74.53.140.153:25 (53 frames, 24 kB) and filtering SMTP greeting through DATA command.

```
21 2009-10-05T02:06:10.661679000-0400 74.53.140.153 25 10.10.1.4 1470 S: 354 Enter message, ending with "." on a line by itself
22 2009-10-05T02:06:10.692743000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
23 2009-10-05T02:06:10.692786000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
24 2009-10-05T02:06:10.692804000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
25 2009-10-05T02:06:10.692823000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Window Full] C: DATA fragment, 1460 bytes
26 2009-10-05T02:06:10.695115000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation need ed)
28 2009-10-05T02:06:10.695623000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation need ed)
29 2009-10-05T02:06:10.696248000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation need ed)
30 2009-10-05T02:06:10.696634000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation need ed)
32 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
33 2009-10-05T02:06:11.104998000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
35 2009-10-05T02:06:11.469759000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
36 2009-10-05T02:06:11.469814000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
38 2009-10-05T02:06:11.494181000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
39 2009-10-05T02:06:11.494199000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
41 2009-10-05T02:06:11.834628000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
42 2009-10-05T02:06:11.834655000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
44 2009-10-05T02:06:11.858316000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
45 2009-10-05T02:06:11.858334000-0400 10.10.1.4 1470 74.53.140.153 25 from: "Gurpartap Singh" <gurpartap@patriots.in>, subject: S MTP, (text/plain) (text/html) (text/plain) (text/plain) (text/plain) | .
52 2009-10-05T02:06:12.248789000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 OK id=1Mugho-0003Dg-Un
54 2009-10-05T02:06:14.763825000-0400 10.10.1.4 1470 74.53.140.153 25 C: QUIT
56 2009-10-05T02:06:15.105467000-0400 74.53.140.153 25 10.10.1.4 1470 S: 221 xc90.websitewelcome.com closing connection
6 2009-10-05T02:06:08.219663000-0400 74.53.140.153 25 10.10.1.4 1470 S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 | We do not authorize the use of this system to transport unsolicited, and/or bulk e-mail.
7 2009-10-05T02:06:08.224809000-0400 10.10.1.4 1470 74.53.140.153 25 C: EHLO GP
9 2009-10-05T02:06:08.566183000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] |
SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP
10 2009-10-05T02:06:08.568729000-0400 10.10.1.4 1470 74.53.140.153 25 C: AUTH LOGIN
11 2009-10-05T02:06:08.911081000-0400 74.53.140.153 25 10.10.1.4 1470 S: 334 VXNlcm5hbWU6
12 2009-10-05T02:06:08.911655000-0400 10.10.1.4 1470 74.53.140.153 25 C: User: Z3VycGFydGFwQHhhdHJpb3RzLmlu
13 2009-10-05T02:06:09.253544000-0400 74.53.140.153 25 10.10.1.4 1470 S: 334 UGFzc3dvcmQ6
14 2009-10-05T02:06:09.254118000-0400 10.10.1.4 1470 74.53.140.153 25 C: Pass: cHVuamF1QDEyMw==
15 2009-10-05T02:06:09.613798000-0400 74.53.140.153 25 10.10.1.4 1470 S: 235 Authentication succeeded
16 2009-10-05T02:06:09.614414000-0400 10.10.1.4 1470 74.53.140.153 25 C: MAIL FROM: <gurpartap@patriots.in>
17 2009-10-05T02:06:09.956765000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 OK
18 2009-10-05T02:06:09.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19 2009-10-05T02:06:10.319708000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 Accepted
20 2009-10-05T02:06:10.320203000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA
```

Figure 5: SMTP Data Fragment Transfer & Extraneous ICMP Notifications — Inspecting 1460-byte payload chunks and identifying ICMP Type 3 Code 4 (Fragmentation Needed) packets.

```
Session Actions Edit View Help
9 2009-10-05T02:06:08.566183000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] |
SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP
10 2009-10-05T02:06:08.568729000-0400 10.10.1.4 1470 74.53.140.153 25 C: AUTH LOGIN
11 2009-10-05T02:06:08.911081000-0400 74.53.140.153 25 10.10.1.4 1470 S: 334 VXNlcm5hbWU6
12 2009-10-05T02:06:08.911655000-0400 10.10.1.4 1470 74.53.140.153 25 C: User: Z3VycGFydGFwQHBhdHJpb3RzLmlu
13 2009-10-05T02:06:08.956765000-0400 74.53.140.153 25 10.10.1.4 1470 S: 334 UGFzc3dvcmQ6
14 2009-10-05T02:06:08.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: Pass: cHVuamFIQDEYMr==
15 2009-10-05T02:06:08.957250000-0400 74.53.140.153 25 10.10.1.4 1470 S: 235 Authentication succeeded
16 2009-10-05T02:06:08.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: MAIL FROM: <gurpartap@patriots.in>
17 2009-10-05T02:06:08.957250000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 OK
18 2009-10-05T02:06:08.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19 2009-10-05T02:06:08.957250000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 Accepted
20 2009-10-05T02:06:08.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA
21 2009-10-05T02:06:08.957250000-0400 74.53.140.153 25 10.10.1.4 1470 S: 354 Enter message, ending with "." on a line by itself
22 2009-10-05T02:06:08.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
23 2009-10-05T02:06:08.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
24 2009-10-05T02:06:08.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
25 2009-10-05T02:06:08.957250000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Window Full] C: DATA fragment, 1460 bytes
26 2009-10-05T02:06:08.957250000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation need
ed)
28 2009-10-05T02:06:08.957250000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation need
ed)
29 2009-10-05T02:06:08.957250000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation need
ed)
30 2009-10-05T02:06:08.957250000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation need
ed)
32 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
33 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
35 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
36 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
38 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
39 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
41 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
42 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
44 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
45 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 from: "Gurpartap Singh" <gurpartap@patriots.in>, subject: S
MTP, (text/plain) (text/html) (text/plain) (text/plain) (text/plain) |
52 2009-10-05T02:06:12.248789000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 OK id=1Mugho-0003Dg-Un
54 2009-10-05T02:06:14.763825000-0400 10.10.1.4 1470 74.53.140.153 25 C: QUIT
56 2009-10-05T02:06:15.105467000-0400 74.53.140.153 25 10.10.1.4 1470 S: 221 xc90.websitewelcome.com closing connection

(kali@kali) ~/SBT-DF203-Lab4
```

Figure 6: Message Finalization, Queue ID Acknowledgment & Session Closure — Reviewing end of DATA, server 250 OK acknowledgment (id=1Mugho-0003Dg-Un), and QUIT command sequence.

```
kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
(kali@kali) ~/SBT-DF203-Lab4
$ cd ~/SBT-DF203-Lab4
tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.p
arameter -e smtp.response.code | tee reports/smtp_commands_responses.tsv
cat reports/smtp_commands_responses.tsv
Warning: program compiled against libxml 215 using older 214
6 2009-10-05T02:06:08.219663000-0400 74.53.140.153 10.10.1.4 220
7 2009-10-05T02:06:08.224809000-0400 10.10.1.4 74.53.140.153 EHLO GP 250
8 2009-10-05T02:06:08.566183000-0400 74.53.140.153 10.10.1.4 250
9 2009-10-05T02:06:08.568729000-0400 10.10.1.4 74.53.140.153 AUTH LOGIN 334
10 2009-10-05T02:06:08.911081000-0400 74.53.140.153 10.10.1.4 334
11 2009-10-05T02:06:08.911655000-0400 10.10.1.4 74.53.140.153 334
12 2009-10-05T02:06:08.956765000-0400 10.10.1.4 74.53.140.153 334
13 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
14 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
15 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
16 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
17 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
18 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
19 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
20 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
21 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
22 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
23 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
24 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
25 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
26 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
27 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
28 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
29 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
30 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
31 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
32 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
33 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
34 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
35 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
36 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
37 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
38 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
39 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
40 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
41 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
42 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
43 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
44 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
45 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
46 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
47 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
48 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
49 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
50 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
51 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
52 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
53 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
54 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
55 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
56 2009-10-05T02:06:08.957250000-0400 10.10.1.4 74.53.140.153 334
```

Figure 7: Clean SMTP Command and Response Transaction Log — Tabulating sequential SMTP commands and numeric response codes via TShark field extractors.

```

(kali㉿kali)-[~/SBT-DF203-Lab4]
$ cd ~/SBT-DF203-Lab4
nano scripts/decode_base64.py

(kali㉿kali)-[~/SBT-DF203-Lab4]
$ python3 scripts/decode_base64.py | tee reports/base64_decoded.txt
cat reports/base64_decoded.txt
== Offline Base64 Decode Results ==
Username: gurpartap@patriots.in
Password: p*****3 (masked for reporting)
== Offline Base64 Decode Results ==
Username: gurpartap@patriots.in
Password: p*****3 (masked for reporting)

(kali㉿kali)-[~/SBT-DF203-Lab4]
$ █

```

Figure 8: Offline Base64 Decoding & Password Masking Script — Executing Python script to decode credentials, masking plaintext password as p*****3 per reporting policy.

```

(kali㉿kali)-[~/SBT-DF203-Lab4]
$ cd ~/SBT-DF203-Lab4
tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 | tee reports/smtp_stream_0.txt
cat reports/smtp_stream_0.txt | head -80
Warning: program compiled against libxml 215 using older 214

=====
Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

9
EHLO GP

137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

18
334 VXNlcm5hbWU6

30
Z3VycGFydGFwQHBhdHJpb3RzLm1u

18
334 UGFzc3dvcmQ6

18
cHVuamFiQDEyMw==

30

```

Figure 9: TCP Stream 0 Follow: Server Banner, EHLO & Authentication Challenge — Reconstructing plaintext stream: Exim 4.69 banner, STARTTLS advertisement, and AUTH LOGIN challenge exchange.

```
kali@kali: ~/JSBT-DF203-Lab4

Session Actions Edit View Help
30
235 Authentication succeeded
36
MAIL FROM: <gurpartap@patriots.in>
8
250 OK
39
RCPT TO: <raj_deol2002in@yahoo.co.in>
14
250 Accepted
6
DATA
56
354 Enter message, ending with "." on a line by itself
1460
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
        .boundary="====_NextPart_000_0004_01CA45B0.095693F0"
X-Mailer: Microsoft Office Outlook 12.0
Thread-Index: AcPfgen9BvjJZEDeR1Kh8i-hUyVo0A=
Content-Language: en-us
x-cr-hashedpuzzle: SoA= AAR2 ADaH BpiO C4G1 D1gW FNB1 FPKR Fn+W HFCP HnYJ JO7s Kum6 KytW LFcI LJUt;1;cgBhAGoAXwBkAGUAbwBsADIAMAAwADIAaQBUEAAeQBhAGgAbwBvAC
4AYwBvAC4AAaQBuaA=;Soshal_v1;7;{CAA37F59-1850-45C7-8540-AA27696B5398};Zw01AHIAcABhAHIAcABhAHAAQABwAGEAdABYAGkAbwB0AHMALgBpAG4A;Mon, 05 Oct 2009 06:06:01 GM
T;UwBNAFQAUAA=
x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398}

This is a multipart message in MIME format.

-----_NextPart_000_0004_01CA45B0.095693F0
Content-Type: multipart/alternative;
        .boundary="====_NextPart_001_0005_01CA45B0.095693F0"
```

Figure 10: TCP Stream 0 Follow: Envelope Fields & MIME Header Reconstruction — Reconstructing MAIL FROM, RCPT TO, DATA command, and multipart/mixed email header boundaries.

```
kali@kali: ~/SBT-DF203-Lab4

Session  Actions  Edit  View  Help

-----=_NextPart_001_0005_01CA45B0.095693F0
Content-Type: text/plain;
.charset="us-ascii"
Content-Transfer-Encoding: 7bit

Hello

I send u smtp pcap file

Find the attachment

GPS

-----=_NextPart_001_0005_01CA45B0.095693F0
Content-Type: text/html;
.charset="us-ascii"
Content-Transfer-Encoding: quoted-printable

<html xmlns:v=3D"urn:schemas-microsoft-com:VML" =
xmlns:o=3D"urn:schemas-microsoft-com:office:office" =
xmlns:w=3D"urn:schemas-microso
1460
ft-com:office:word" =
xmlns:m=3D"http://schemas.microsoft.com/office/2004/12/omml" =
xmlns=3D"http://www.w3.org/TR/REC-html40">

<head>
<META HTTP-EQUIV=3D"Content-Type" CONTENT=3D"text/html; =
charset=3Dus-ascii">
<meta name=3DGenerator content=3D"Microsoft Word 12 (filtered medium)">
<style>
<!--
/* Font Definitions */
@font-face
.{font-family:"Cambria Math";
.panose-1:2 4 5 3 5 4 6 3 2 4;}
@font-face
.{font-family:Calibri;
```

Figure 11: TCP Stream 0 Follow: Plaintext Message Body & HTML Part Start — Displaying text/plain message body ("Hello I send u smtp pcap file...") and opening of text/html part.

```
kali@kali: ~/SBT-DF203-Lab4

Session Actions Edit View Help

.{font-family:Calibri;
.panose-1:2 15 5 2 2 2 4 3 2 4;}
/* Style Definitions */
p.MsoNormal, li.MsoNormal, div.MsoNormal
.{margin:0in;
.margin-bottom:.0001pt;
.font-size:11.0pt;
.font-family:"Calibri","sans-serif";}
a:link, span.MsoHyperlink
.{mso-style-priority:99;
.color:blue;
.text-decoration:underline;}
a:visited, span.MsoHyperlinkFollowed
.{mso-style-priority:99;
.color:purple;
.text-decoration:underline;}
span.EmailStyle17
.{mso-style-type:personal-compose;
.font-family:"Calibri","sans-serif";
.color:windowtext;}
..MsoChpDefault
.{mso-style-type:export-only;}
@page Section1
.{size:8.5in 11.0in;
.margin:1.0in 1.0in 1.0in 1.0in;}
div.Section1
.{page:Section1;}
→
</style>
<!--[if gte mso 9]><xml>
  <o:shapedefaults v:ext=3D"edit" spidmax=3D"1026" />
</xml><![endif]>→<!--[if gte mso 9]><xml>
  <o:shapelayout v:ext=3D"edit">
    <o:idmap v:ext=3D"edit" data=3D"1" />
  </o:shapelayout></xml><![endif]>→
</head>

<body lang=3DEN-US link=3Dblue vlink=3Dpurple>

<div class=3DSection1>
1460
```

Figure 12: TCP Stream 0 Follow: Word 12 HTML Formatting & Font Styles — Dissecting Microsoft Word 12 HTML document styles, Cambria Math, and Calibri typography markup.

```
kali@kali: ~/SBT-DF203-Lab4

Session  Actions  Edit  View  Help

<p class=3DMsoNormal>Hello<o:p></o:p></p>

<p class=3DMsoNormal><o:p>&nbsp;</o:p></p>

<p class=3DMsoNormal>I send u smtp pcap file <o:p></o:p></p>

<p class=3DMsoNormal>Find the attachment<o:p></o:p></p>

<p class=3DMsoNormal><o:p>&nbsp;</o:p></p>

<p class=3DMsoNormal>GPS<o:p></o:p></p>

</div>

</body>

</html>

-----=_NextPart_001_0005_01CA45B0.095693F0--

-----=_NextPart_000_0004_01CA45B0.095693F0
Content-Type: text/plain;
.name="NEWS.txt"
Content-Transfer-Encoding: quoted-printable
Content-Disposition: attachment;
.filename="NEWS.txt"

Version 4.9.9.1
* Many bug fixes
* Improved editor

Version 4.9.9.0
* Support for latest Mingw compiler system builds
* Bug fixes

Version 4.9.8.9
* New code tooltip display
* Improved Indent/Unindent and Remove Comment
* Improved automatic indent
* Added support for the "interface" keyword
* WebUpdate should now report installation problems from PackMan
* New splash screen and association icons
```

Figure 13: TCP Stream 0 Follow: HTML Body Conclusion & Attachment Header — End of HTML segment and boundary marker opening attachment NEWS.txt (Dev-C++ history).

```
kali@kali: ~/SBT-DF203-Lab4

Session Actions Edit View Help

* New splash screen and association icons
* Improved installer
* Many bug fixes

Version 4.9.8.7
* Added support for GCC > 3.2
* Debug variables are now resent during next debug session
* Watched Variables not in correct context are now kept and updated when =
it is needed
* Added new compiler/linker options:=20
  - Strip executable
  - Generate instructions for a specific machine (i386, i486, i586, =
i686, pentium, pentium-mmx, pentiumpro, pentium2, pentium3, pentium4,=20
k6, k6-2, k6-3
1460
, athlon, athlon-tbird, athlon-4, athlon-xp, =
athlon-mp, winchip-c6, winchip2, k8, c3 and c3-2)
  - Enable use of processor specific built-in functions (mmmx, sse, =
sse2, pni, 3dnow)
* "Default" button in Compiler Options is back
* Error messages parsing improved
* Bug fixes

Version 4.9.8.5
* Added the possibility to modify the value of a variable during =
debugging (right click on a watch variable and select "Modify value")
* During Dev-C++ First Time COnfiguration window, users can now choose =
between using or not class browser and code completion features.
* Many bug fixes

Version 4.9.8.4
* Added the possibility to specify an include directory for the code =
completion cache to be created at Dev-C++ first startup
* Improved code completion cache
* WebUpdate will now backup downloaded DevPaks in Dev-C++\Packages =
directory, and Dev-C++ executable in devcpp.exe.BACKUP
* Big speed up in function parameters listing while editing
* Bug fixes

Version 4.9.8.3
* On Dev-C++ first time configuration dialog, a code completion cache of =
all the standard=20
include files can now be generated.
```

Figure 14: Attachment Inspection: NEWS.txt (Versions 4.9.8.7 to 4.9.9.1) — Carving text attachment: reviewing release notes for Dev-C++ v4.9.8.7 through v4.9.9.1.

```
Session  Actions  Edit  View  Help
include files can now be generated.
* Improved WebUpdate module
* Many bug fixes

Version 4.9.8.2
* New debug feature for DLLs: attach to a running process
* New project option: Use custom Makefile.=20
* New WebUpdater module.
* Allow user to specify an alternate configuration file in Environment =
Options=20
  (still can be overridden by using "-c
1420
" command line parameter).
* Lots of bug fixes.

Version 4.9.8.1
* When creating a DLL, the created static lib respects now the =
project-defined output directory

Version 4.9.8.0
* Changed position of compiler/linker parameters in Project Options.
* Improved help file
* Bug fixes

Version 4.9.7.9
* Resource errors are now reported in the Resource sheet
* Many bug fixes

Version 4.9.7.8
* Made whole bottom report control floating instead of only debug =
output.
* Many bug fixes

Version 4.9.7.7
* Printing settings are now saved
* New environment options : "watch variable under mouse" and "Report =
watch errors"
* Bug fixes

Version 4.9.7.6
* Debug variable browser
* Added possibility to include in a Template the Project's directories =
(include, libs and ressources)
```

Figure 15: Attachment Inspection: NEWS.txt (Versions 4.9.7.6 to 4.9.8.2) — Carving text attachment: reviewing release notes for Dev-C++ v4.9.7.6 through v4.9.8.2.

```
Session Actions Edit View Help
(include, libs and ressources)
* Changed tint of Class browser pictures colors to match the New Look =
style
* Bug fixes

Version 4.9.7.5
* Bug fixes

Version 4.9.7.4
* When compiling with debugging symbols, an extra definition is passed =
to the
  compiler: -D__DEBUG__
* Each project creates a <project_name>_private.h file containing =
version
  information definitions
* When compiling the current file only, no dependency checks are =
performed
* ~300% Speed-up in class parser
* Added "External programs" in Tools/Environment Options (for units =
"Open with")
* Added "Open with" in project units context menu
* Added
1452
  "Classes" toolbar
* Fixed pre-compilation dependency checks to work correctly
* Added new file menu entry: Save Project As
* Bug-fix for double quotes in devcpp.cfg file read by vUpdate
* Other bug fixes

Version 4.9.7.3
* When adding debugging symbols on request, remove "-s" option from =
linker
* Compiling progress window
* Environment options : "Show progress window" and "Auto-close progress =
window"
* Bug fixes

Version 4.9.7.2
* Bug fixes

Version 4.9.7.1
* "Build priority" per-unit
* "Include file in linking process" per-unit
```

Figure 16: Attachment Inspection: NEWS.txt (Versions 4.9.7.1 to 4.9.7.5) — Carving text attachment: reviewing release notes for Dev-C++ v4.9.7.1 through v4.9.7.5.

Session Actions Edit View Help

- * "Include file in linking `process`" per-unit
- * New feature: compile current file only
- * Separated C++ compiler options from C compiler options in Makefile = (see bug report #654744)
- * Separated C++ include dirs from C include dirs in Makefile (see bug = report #654744)
- * Necessary UI changes in Project Options
- * Added display of project filename, project output and a summary of the = project files in Project Options General tab.
- * Fixed the "compiler-dirs-with-spaces" bug that crept-in in 4.9.7.0
- * Multi-select files in project-view (when "double-click to open" is = configured in Environment Settings)
- * Resource files are treated as ordinary files now
- * Updates in "Project Options/Files" code
- * MSVC import now creates the folders structure of the original VC = project
- * Bug fixes

Version 4.9.7.0

- * Allow customizing of per-unit compile command in projects
- * Added two new macros: `<DATE>` and `<DATETIME>`
- * A
1452
added support for macros in the "default source code" (Tools/Editor = Options/Code)
- * Separated layout info from project file. It is now kept in a different = file
(the same filename as the project's but with extension ".layout"). If = you
have your project under CVS control, you 'll know why this had to = happen ...
- * Compiler settings per-project
- * Compiler set per-project
- * Implemented new compiler settings framework
- * "Compile as C++" per-unit
- * "Include file in compilation process" per-unit
- * Project version info (creates the relevant VERSIONINFO struct in the = private
resource)
- * Support XP Themes (creates the CommonControls 6.0 manifest file and = includes
it in the private resource)
- * Added CVS "login" and "logout" commands

Figure 17: Attachment Inspection: NEWS.txt (Version 4.9.7.0) — Carving text attachment: reviewing release notes for Dev-C++ v4.9.7.0 and CVS integration.

```

Session Actions Edit View Help
* Added CVS "login" and "logout" commands
* Project manager and debugging window (in Debug tab) can now be =
transnformed into floating windows.
* Added "Add Library" button in Project Options
* Bug fixes

Version 4.9.6.9
* Implemented search in help files for the word at cursor (context =
sensitive help)
* Implemented "compiler sets" infrastructure to switch between different =
compilers easily (e.g. gcc-2.95 and gcc-3.2)
* Added "Files" tab in CVS form to allow selection of more than one file =
for
the requested CVS action
=20
Version 4.9.6.8
* support for DLL application hosting, for debugging and executing DLLs =
under Dev-C++.
* New class browser option: "Show inherited members"
* Added sup
1452
port for the '::' member access operator in code-completion
* Added *working* function arguments hint
* Added bracket highlighting. When the caret is on a bracket, that =
bracket and
its counterpart are highlighted
* Nested folders in project view

Version 4.9.6.7
* XP Theme support
* Added CVS commands "Add" and "Remove"
* Added configuration option for "Templates Directory" in "Environment =
Options"
* Code-completion updates
* Bug fixes

Version 4.9.6.6
* Editor colors are initialized properly on Dev-C++ first-run
* Added doxygen-style comments in NewClass, NewMemberFunction and =
NewMemberVariable wizards
* Added file's date/time stamp in File/Properties window
* Current windows listing in Window menu
* Bug fixes

```

Figure 18: Attachment Inspection: *NEWS.txt* (Versions 4.9.6.6 to 4.9.6.9) — Carving text attachment: reviewing release notes for Dev-C++ v4.9.6.6 through v4.9.6.9.

```
Session Actions Edit View Help
* Bug fixes

Version 4.9.6.5
* CVS support
* Window list (in Window menu)
* bug fixes

version 4.9.6.4
* added ENTER key for opening file in project browser, DEL to delete =
from the project.
* bug fixes

version 4.9.6.3
* Bug fixes

version 4.9.6.2
* Bug fixes

version 4.9.6.1
* New "Abort compilation" button
* Bug fixes
* Now checks for vRoach existence when sending a crash report

Version 4.9.5.5
* New option in Editor Options: Show editor hints. User can disable the =
hints
  displayed in the editor when the mouse moves over a word. Since this =
was the
  cause of many errors (although it should be fixed by now), we are =
giving the
  user the option to disable this featu
1452
re.
* New option in Editor Options (code-completion): Use code-completion =
cache.
  Well, it adds caching to code-completion. Depending on the cache size,
  the program may take a bit longer to start-up, but provides very fast
  code-completion and the user has all the commands (belonging to the =
files
  he added in the cache) at his fingertips. If, for example, the user =
adds
  "windows.h", he gets all the WinAPI! If he adds "wx/wx.h", he gets all =
of
```

Figure 19: Attachment Inspection: NEWS.txt (Versions 4.9.5.5 to 4.9.6.5) — Carving text attachment: reviewing release notes for Dev-C++ v4.9.5.5 through v4.9.6.5.

```

of
wxWindows! You get the picture ...
* Removed "Only show classes from current file" option in class browser =
settings.
  It used to be a checkbox, allowing only two states (on or off), but =
there is
  a third relevant option now: "Project classes" so it didn't fit the =
purpose ...
  The user can define this in the class browser's context menu under =
"View mode".
* Fixed the dreaded "Clock skew detected" compiler warning!
* Fixed many class browser bugs, including some that had to do with =
class folders.

Version 4.9.5.4
* Under NT, 2000 and XP, user application data directory will be used to =
store config files (i.e : C:\Documents and Settings\Username\Local =
Settings\Application Data)

Version 4.9.5.3
* Added ExceptionsAnalyzer. If the devcpp.map file is in the devcpp.exe =
directory
  then we even get a stack trace in the bug report!
* Added new WebUpdate module (inactive temporarily).
* Added new code for code-completion caching of fi
1452
les (disabled - work =
in progress).

Version 4.9.5.2
* Added new option in class-browser: Use colors
  (available when right-clicking the class-browser
  and selecting "View mode").
* Dev-C++ now traps access violation of your programs (and of itself too =
;)

Version 4.9.5.1
* Implemented the "File/Export/Project to HTML" function.
* Added "Tip of the day" system.
* When running a source file in explorer, don't spawn new instance.
  Instead open the file in an already launched Dev-C++.
* Class-parser speed-up (50% to 85% improvement timed!!!)
* Many code-completion updates. Now takes into account context,

```

Figure 20: Attachment Inspection: NEWS.txt (Versions 4.9.5.1 to 4.9.5.4) — Carving text attachment: reviewing release notes for Dev-C++ v4.9.5.1 through v4.9.5.4.

```

Session  Actions  Edit  View  Help
* Many code-completion updates. Now takes into account context,
  class inheritance and visibility (shows items only from files
  #included directly or indirectly)!
* Caching of result set of code-completion for speed-up.
* New option "Execution/Parameters" (and "Debug/Parameters").

Version 4.9.5.0 (5.0 beta 5):
* CPU Window (still in development)
* ToDo list
* Backtrace in debugging
* Run to cursor
* Folders in Project and Class Browser
* Send custom commands to GDB
* Makefile can now be customized.
* Modified the behaviour of the -c param :=20
  -c <config file directory>
* Saving of custom syntax parameter group
* Possibility of changing compilers and tools filename.
* Many bug fixes

Version 4.9.4.1 (5.0 beta 4.1):
* back to gcc 2.95.3
* Profiling support
* new update/packages checker (vUpdate)
* Lots of bugfixes

-----=_NextPart_000_00
29
04_01CA45B0.095693F0--
.

      28
250 OK id=1Mugho-0003Dg-Un

6
QUIT

      48
221 xc90.websitewelcome.com closing connection

```

Figure 21: Attachment Inspection & Stream Conclusion: NEWS.txt (v4.9.4.1 to v4.9.5.0) — Concluding attachment carving with v4.9.4.1 release notes, MIME boundary closing, queue ID 250 OK, and QUIT sequence.

```

Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab4]
└─$ grep -E1 '^([Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|Content-Transfer-Encoding|User-Agent|X-Mailer]):' reports/sntp_stream_0.txt | tee
reports/message_headers.txt
cat reports/message_headers.txt
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Transfer-Encoding: 7bit
Content-Type: text/html;
Content-Transfer-Encoding: quoted-printable
Content-Type: text/plain;
Content-Transfer-Encoding: quoted-printable
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Transfer-Encoding: 7bit
Content-Type: text/html;
Content-Transfer-Encoding: quoted-printable
Content-Type: text/plain;
Content-Transfer-Encoding: quoted-printable

(kali@kali)-[~/SBT-DF203-Lab4]
└─$ cp reports/sntp_stream_0.txt reports/reconstructed_email_redacted.txt
acho "==== REDACTION NOTE: Passwords and credentials masked per ICDA policy ===" | tee -a reports/reconstructed_email_redacted.txt
==== REDACTION NOTE: Passwords and credentials masked per ICDA policy ===

(kali@kali)-[~/SBT-DF203-Lab4]
└─$

```

Figure 22: Extracted Email Headers & Redacted File Archive — Grep extraction of message headers and generating working copy reconstructed_email_redacted.txt.

```

(kali@kali)-[~/SBT-DF203-Lab4]
└─$ cd ~/SBT-DF203-Lab4
tshark -r working/sntp_working.pcap -Y 'smtp' -T fields -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport -e ip.dst -e tcp
.dstport -e tcp.stream | tee reports/sntp_network_metadata.tsv
cat reports/sntp_network_metadata.tsv | head -20
Warning: program compiled against libxml 215 using older 214
6      1254722768.219663000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  74.53.140.153  25  10.10.1.4  1470  0
7      1254722768.224809000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
9      1254722768.566183000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  74.53.140.153  25  10.10.1.4  1470  0
10     1254722768.568729000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
11     1254722768.911081000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  74.53.140.153  25  10.10.1.4  1470  0
12     1254722768.911655000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
13     1254722769.253544000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  74.53.140.153  25  10.10.1.4  1470  0
14     1254722769.254118000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
15     1254722769.613798000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  74.53.140.153  25  10.10.1.4  1470  0
16     1254722769.614414000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
17     1254722769.956765000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  74.53.140.153  25  10.10.1.4  1470  0
18     1254722769.957250000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
19     1254722770.319708000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  74.53.140.153  25  10.10.1.4  1470  0
20     1254722770.320203000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
21     1254722770.661679000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  74.53.140.153  25  10.10.1.4  1470  0
22     1254722770.692743000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
23     1254722770.692786000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
24     1254722770.692804000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
25     1254722770.692823000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
26     1254722770.695115000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  192.168.1.1,10.10.1.4  1470  10.10.1.4,74.53.140.153  25  0
28     1254722770.695623000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  192.168.1.1,10.10.1.4  1470  10.10.1.4,74.53.140.153  25  0
29     1254722770.696248000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  192.168.1.1,10.10.1.4  1470  10.10.1.4,74.53.140.153  25  0
30     1254722770.696631000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  192.168.1.1,10.10.1.4  1470  10.10.1.4,74.53.140.153  25  0
32     1254722771.104072000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
33     1254722771.104098000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
35     1254722771.469759000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
36     1254722771.469814000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
38     1254722771.494818000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
39     1254722771.494199000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
41     1254722771.834628000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
42     1254722771.834655000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
44     1254722771.858316000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
45     1254722771.858334000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
52     1254722772.248789000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  74.53.140.153  25  10.10.1.4  1470  0
54     1254722774.763825000  00:e0:1c:3c:17:c2  00:1f:33:d9:81:60  10.10.1.4  1470  74.53.140.153  25  0
56     1254722775.105467000  00:1f:33:d9:81:60  00:e0:1c:3c:17:c2  74.53.140.153  25  10.10.1.4  1470  0

```

Figure 23: Complete SMTP Layer 2 & Layer 3 Metadata Dissection — Tabulating frame timestamps, Ethernet MAC addresses, IP addresses, and TCP port mappings.

```

(kali@kali)-[~/SBT-DF203-Lab4]
$ cd ~/SBT-DF203-Lab4
echo "=== STARTTLS Check ===" | tee reports/tls_assessment.txt
echo "Server advertised STARTTLS:" | tee -a reports/tls_assessment.txt
tshark -r working/smtp_working.pcap -Y 'smtp contains "STARTTLS"' -T fields -e frame.number -e _ws.col.Info | tee -a reports/tls_assessment.txt
echo "" | tee -a reports/tls_assessment.txt
echo "Client issued STARTTLS command:" | tee -a reports/tls_assessment.txt
tshark -r working/smtp_working.pcap -Y 'smtp.req.command == "STARTTLS"' -T fields -e frame.number | tee -a reports/tls_assessment.txt
echo "(If empty above, STARTTLS was NOT used)" | tee -a reports/tls_assessment.txt
cat reports/tls_assessment.txt
=== STARTTLS Check ===
Server advertised STARTTLS:
Warning: program compiled against libxml 215 using older 214
9      S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP

Client issued STARTTLS command:
Warning: program compiled against libxml 215 using older 214
(If empty above, STARTTLS was NOT used)
=== STARTTLS Check ===
Server advertised STARTTLS:
9      S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP

Client issued STARTTLS command:
(If empty above, STARTTLS was NOT used)

(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'tls' | wc -l | tee -a reports/tls_assessment.txt
echo "TLS packets found (0 = no encryption)" | tee -a reports/tls_assessment.txt
Warning: program compiled against libxml 215 using older 214
0
TLS packets found (0 = no encryption)

(kali@kali)-[~/SBT-DF203-Lab4]
$ >....
 334 Username challenge / 334 Password challenge
 235 Authentication succeeded
 MAIL FROM: <gurpartapatriots.in>
 RCPT TO: <raj_deol2002in@yahoo.co.in>
 DATA
 354 Enter message
 250 OK id=1Mugho-0003Dg-Un

```

Figure 24: STARTTLS Assessment & TLS Packet Count Verification — Script verification proving server advertised STARTTLS, client issued 0 STARTTLS commands, and 0 TLS packets were transmitted.

9. Conclusions

This lab successfully demonstrated the forensic analysis of a plaintext SMTP session. The capture revealed a complete email exchange including credentials, message headers, body, and attachment. The absence of STARTTLS usage resulted in full plaintext exposure — a critical finding that highlights the importance of encryption in email communications. Base64 encoding provided no confidentiality. All evidence was preserved with matching SHA-256 hashes.

10. Recommendations

- **Enforce Mandatory TLS (STARTTLS):** Mail Transfer Agents (MTAs) and Mail Submission Agents (MSAs) should mandate opportunistic or enforced TLS (e.g., SMTP over Port 587/465) to block cleartext transmission of credentials and email content.
- **Deprecate Cleartext Authentication Over Port 25:** Port 25 should be restricted purely to MTA-to-MTA server relay. Client mail submissions must use port 587 with mandatory authentication wrapped within TLS.
- **Differentiate Base64 Encoding from Encryption:** Forensic awareness training must emphasize that Base64 is a data format encoding, providing zero cryptographic confidentiality.

11. Appendix – Commands and Hashes

```
# Evidence Hash Verification
sha256sum evidence/smtp.pcap working/smtp_working.pcap

# Capture File Statistics
capinfos evidence/smtp.pcap

# TCP Conversation & Stream Reconstruction
tshark -r working/smtp_working.pcap -q -z conv,tcp
tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0

# Filter SMTP Commands & Responses
tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields ...

# Offline Credential Decoding
python3 scripts/decode_base64.py
```

Evidence Integrity: Cryptographic Hash Digest Preservation:

- **Original Evidence Hash:** 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
(evidence/smtp.pcap)
- **Working Copy Hash:** 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
(working/smtp_working.pcap)

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.

Student Signature: Almustapha Yusuf

Date: 14/09/2026